SYSTEM ARCHITECTURE — MODULAR ETL · V1.0

Oncology Clinical Trials *Analytics Platform*

End-to-end data engineering pipeline — from ClinicalTrials.gov REST ingestion through incremental watermark-based ETL, MySQL persistence, and Streamlit + Plotly interactive analytics.

- Production Architecture
- 10 Architectural Layers
- 6 Plotly Chart Types
- 2 MySQL Tables

01 EXTERNAL DATA SOURCE

CLINICALTRIALS.GOV · NIH

REST API — JSON Data Source

REST / HTTPS | JSON Responses | Paginated | last_update_posted | Oncology Trials

↓HTTP · JSON PAYLOAD

02 INGESTION LAYER

⚙ PYTHON MODULE

api_client.py

- Sends HTTP GET requests to ClinicalTrials.gov endpoint
- Manages automatic pagination across all result pages
- Configurable filters: `condition` · `status` · `date range`
- Returns raw JSON payload to pipeline orchestrator

↓RAW JSON RECORDS

03–05 CORE ETL ENGINE

CORE ENGINE

ORCHESTRATOR

run_pipeline.py

- Retrieves watermark from `etl_run_logs`
- Sequences extract → transform → load
- Triggers transformation on raw data
- Writes run metadata on completion

TRANSFORM

transform.py

- Flattens deeply nested JSON
- Extracts structured analytical fields
- Parses & normalises date formats
- Handles nulls, type casting, defaults

INCREMENTAL

Watermark Logic

- Compare `last_update_posted` vs watermark

- Load only new / updated records
- Skip entirely unchanged records
- Advance watermark on success

↓CLEANED, FILTERED RECORDS

06 LOAD LAYER

UPSERT STRATEGY

load.py

- Batch INSERT into MySQL `clinical_trials` table
- `ON DUPLICATE KEY UPDATE` — safe idempotent upsert
- Deduplication enforced on `nct_id` primary key
- Transaction commit on success · full rollback on failure

↓PERSISTED TO STORAGE

07–08 DATABASE LAYER & MONITORING

MySQL Database

ANALYTICAL · INDEXED · PERSISTENT

T clinical_trials

nct_id	PK
title	TEXT
status · phase	VARCHAR
sponsor · condition	VARCHAR
start_date · end_date	DATE
last_update_posted	DATE
country · enrollment	VARCHAR / INT

✓ Indexed Analytical Columns

T etl_run_logs

run_id	PK
start_time	DATETIME
end_time	DATETIME
duration_ms	INT
status	ENUM
records_inserted	INT
records_updated	INT
error_msg	TEXT

OBSERVABILITY

ETL Run Monitoring

- Persist metadata for every pipeline run
- Track duration in milliseconds
- Status: `SUCCESS` / `FAILED` / `PARTIAL`
- Record counts: inserted vs updated
- Watermark timestamp advancement
- Full error message capture on failure
- Run history queryable from dashboard

↓SQL QUERIES VIA SQLALCHEMY

09 ANALYTICS LAYER

Streamlit Dashboard Application

MULTI-PAGE · CACHED · INTERACTIVE FILTERS

@st.cache_data | SQLAlchemy Pool | Sidebar Filters | Phase / Status / Year / Country | ETL Monitor Page | Analytics Page

↓PLOTLY FIGURES RENDERED

10 VISUALIZATION LAYER — PLOTLY INTERACTIVE CHARTS

KPI Cards

Total trials · Active & recruiting counts · Phase distribution summary metrics

Distribution Charts

Phase breakdown · Status distribution · Sponsor type — Plotly pie & bar charts

Yearly Trends

Trials started per year · Enrollment over time · Completion rates — line & bar

Geographic Map

Plotly choropleth · Trials by country · Color-scaled density · Interactive tooltips

Heatmap

Phase × Status correlation matrix · Density colour scale · Hover cell details

ETL Monitor Dashboard

Pipeline run history · Duration trend · Success/failure rates · Watermark log table